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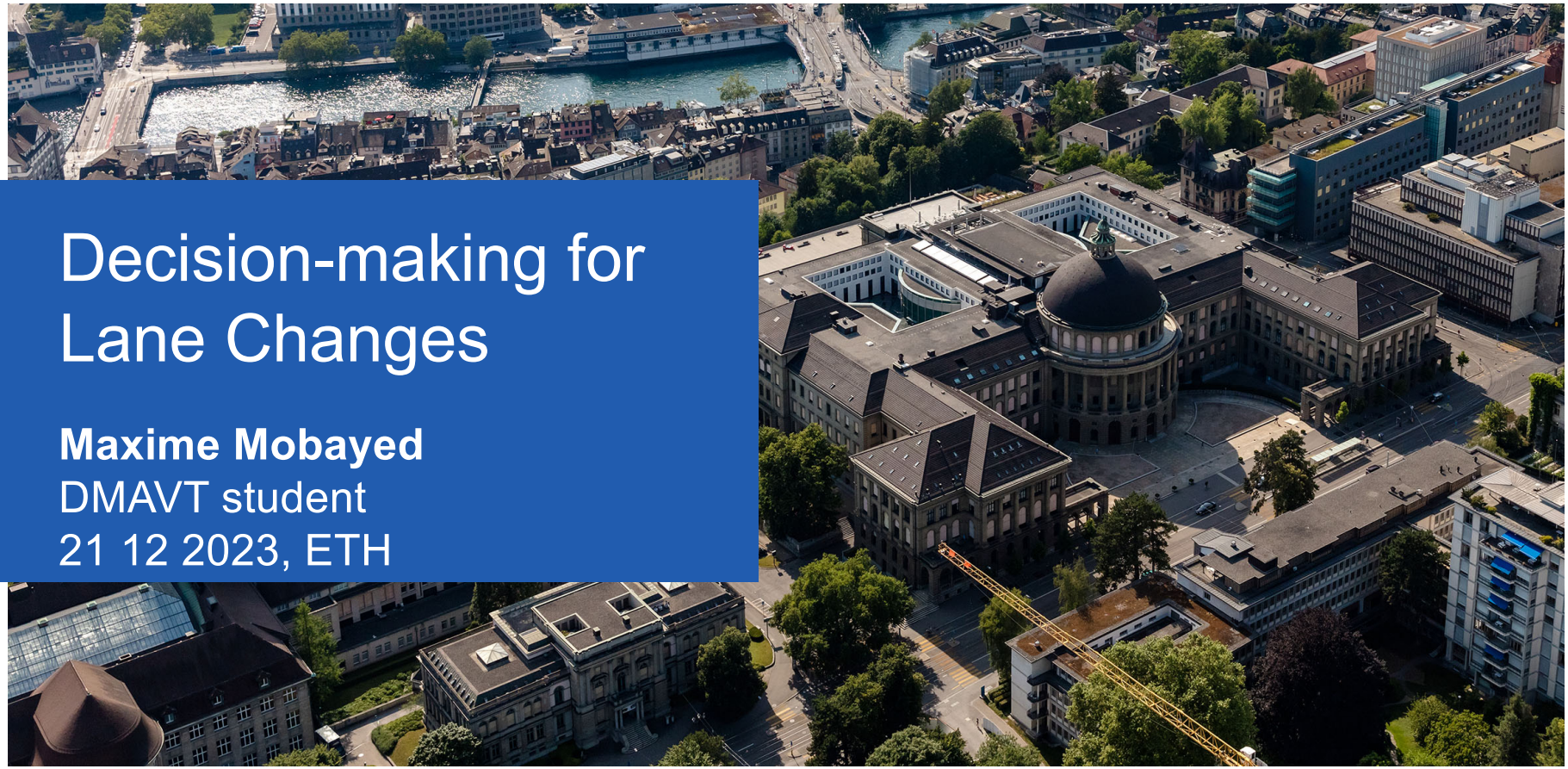

Decision-making for Lane Changes

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21 12 2023, ETH

IDSC

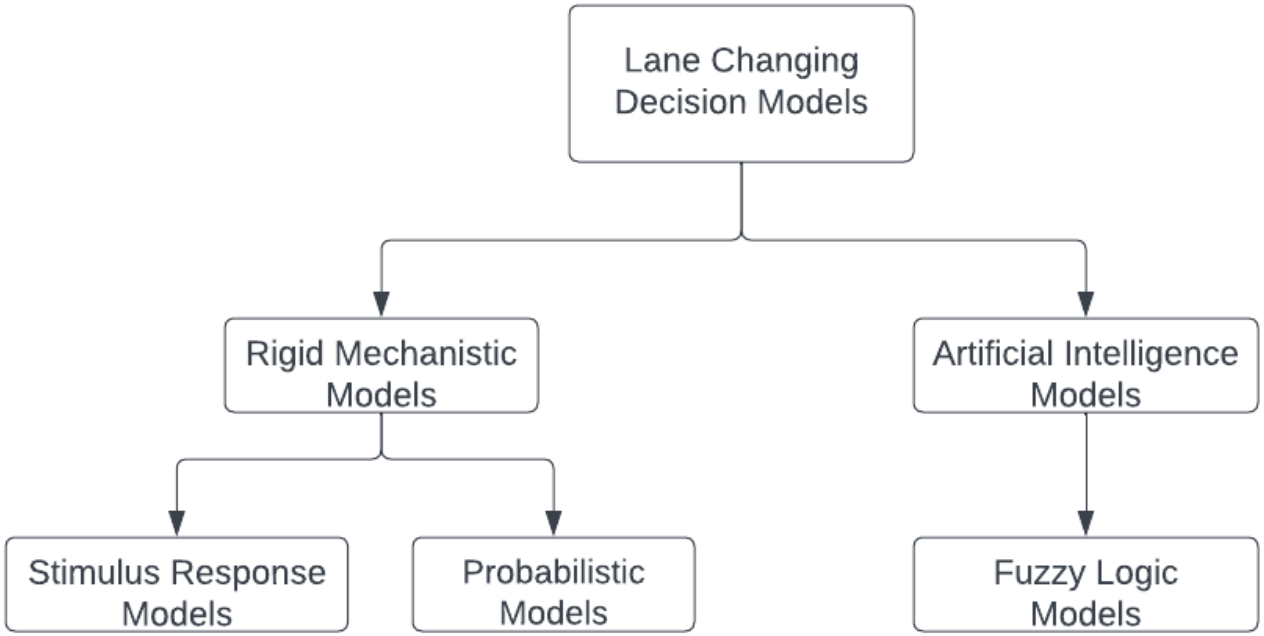
D MAVT

Why is this topic relevant today?

- AV is becoming a reality, it is joining the current traffic flow.
- We need to set rules for AVs, so that Human drivers and AVs can coexist.

Impact

- Macroscopic (average traffic speed, density, etc.) and microscopic (trajectory of the car, distance headways, etc.) characteristic of traffic flows
- Transport planning and development of traffic management policies (number of connection, number of lanes, maximum allowed speed, etc.)



Rigid Mechanistic model

- Explanatory and dependent variables
- Do not incorporate uncertainties driver behavior → well defined probabilities

Artificial Intelligence model

- Imprecise perception of surrounding traffic
- Uncertainty through the subjective perception

Table of content

1. Rigid Mechanistic model
    1. Stimulus response
        1. Gibbs
    2. Discrete choice models
        1. Ahmed
        2. Toldeo
        3. Wen
2. Artificial Intelligence model
    1. Fuzzy logic
        1. McDonald
        2. David and Söffker
        3. Tang
3. Conclusion

Stimulus Response

- Stimulus = what the driver is able to perceive from the environment
  - Example: relative speed, speed, signals, pavement marking view of the lane, sound of cars, etc.
- The behavior of the driver depends on the observations it makes
  - Example: aggressive or unfamiliar drivers they rely on stimulus from the environment to respond. While regular commuters such as taxi drivers do not rely on the environment stimulus response, and rather rely on their knowledge.
- Explanatory & dependent variable various papers with different variable definition

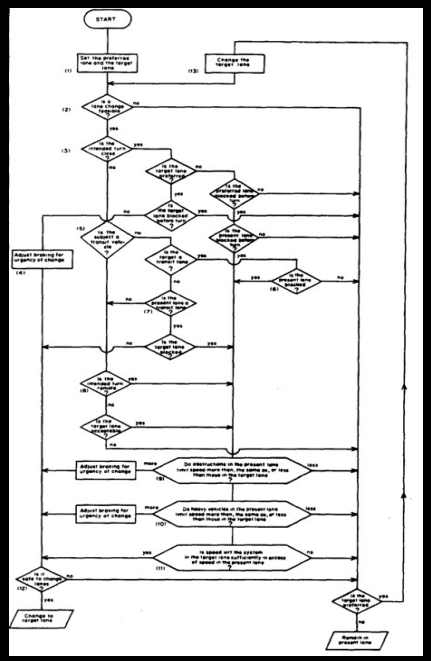
Gipps, 1986 [4]

- Gipps model is developed for highway or urban scenarios
- Behavior while entering, traveling along, and exiting off a section road
- Explanatory variables:
  - Maximum subject vehicle safe speed,
  - Lead gap
  - Subject vehicles drivers estimation of front vehicles drivers brake.
- Average drivers 3 Questions:
  - Is it possible to change lanes?
  - Is it necessary to change lanes?
  - Is it desirable to change lanes?
- Drivers goal:
  - Safety
  - Comfort
  - Time of arrival
- Three patterns to portray a drivers behavior:
  - Far priority →desired speed
  - Middle priority → correct lane over speed
  - Close priority → appropriate lane for his turn

- Model is a combination with car following model → which sets limitations on drivers breaking rate[3]:

$$v_{n(t+T)} = bnT + [b^2{}_n T^2 - bn\left(2\{x_{n_1}(t) - sn_{_1_}xn(t)\} - vn(t)T - \frac{v_{n_1}(t)^2}{b}\right)]^{1/2}$$

- The decision for Lane change are based on a 13 question process. Questions are subjective or objective



Decision making flowchart [4]

Probabilistic model

- Probabilistic model, is based on
  - Independent stages
    - Decide on Lane change
    - Select Target Lane
    - Gap Acceptance
  - Explanatory variables
    - Exit distance
    - Relative speed
    - Presence of heavy vehicles
  - Probability function
    - maximum utility gain

Ahmed, 1999[1]

- 3 decision stages:
  - Consider a lane change.
  - Choice of Target Lane (TL).
  - Gap acceptance
- MLC = Mandatory Lane Change
  - Influential variables:
    - remaining distance to the point at which lane change must be completed, delay, number of lanes to cross, subject vehicle is heavy, etc.
- <u>MLC</u> = “NOT Mandatory Lane Change”
- DLC = Discretionary Lane Change
  - Unsatisfied with current lane

- Lane selection
- Gap Acceptance[1]: Evaluates lead & lag gap in TL

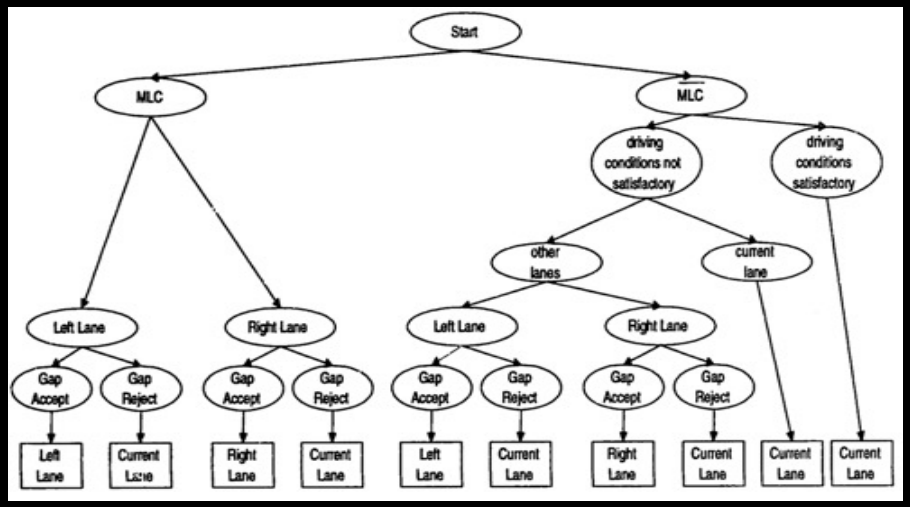
$$P\left(G_n^{cr,lead}(t) > G_n^{cr,lead}(t)\middle|v_n\right)P\left(G_n^{cr,lag}(t) > G_n^{cr,TL,cr}(t)\middle|v_n\right)$$
$$= \phi\left[\frac{ln(G_n^{lead}(t)) - (X_n^{lead}(t)\beta^{lead} + \alpha^{lead}v_n)}{\sigma_{\varepsilon lead}}\right] \text{ x } \phi\left[\frac{ln(G_n^{lag}(t)) - (X_n^{lag}(t)\beta^{lag} + \alpha^{lag}v_n)}{\sigma_{\varepsilon lag}}\right]$$

- Critical gap size[1]

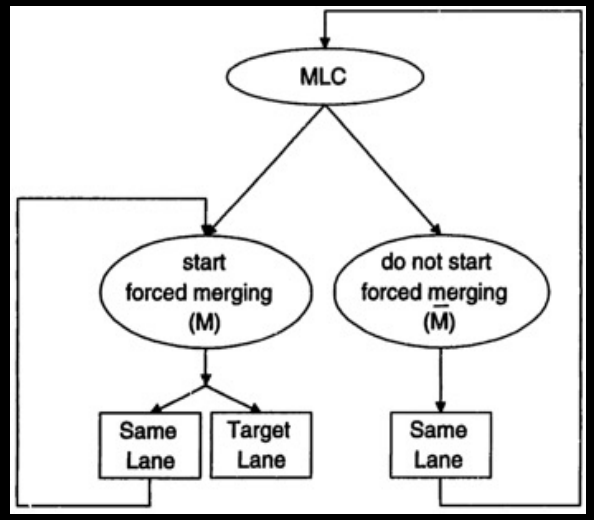
$$G_n^{cr,g}(t) = \exp(X_n^g(t)\beta^g + \alpha^g v_n + \epsilon_n^g(t)$$

- Forced Merging Model

MLC Flowchart [1]



Forced Merging Flowchart[1]



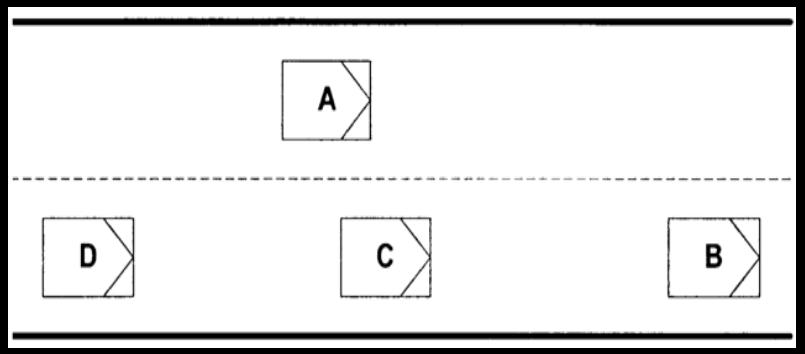
Probability of Lane Change[1]

$$P_t(LC|v_n) = \frac{1}{1 + \exp(-X_n^{LC}(t)\beta^{LC} - \alpha^{LC} v_n)}$$

Toledo, 2003[7]

- Criticizes the limitations of existing models
- Toledo classifies the considerations broadly into four categories:
  - Neighborhood variables include factors like relative positions, speeds, proximity to other vehicles, and police presence.
  - Path plan variables involve how a driver chooses lanes and performs lane changes, affecting speed and acceleration choices.
  - Network knowledge and experience reflect a driver's experience with the transportation system.
  - Driving style and capabilities are individual characteristics that can influence driving behavior.

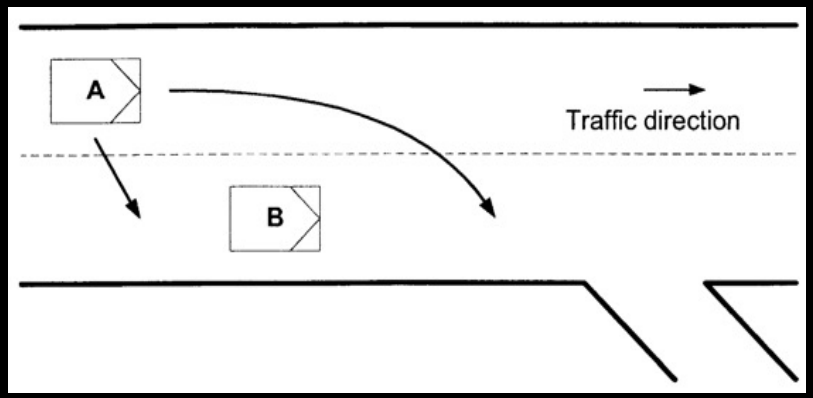
- Short-term Goal and Plan[7]



- This model integrates DLC and MLC into a single utility function[7] for each TL
  - Include the choice of changing left or right

$$U_n^{CL}(t) = X_n^{CL}(t)\beta^{CL} + \alpha^{CL}v_n + \varepsilon_n^{CL}(t)$$
$$U_n^{RL}(t) = X_n^{RL}(t)\beta^{RL} + \alpha^{RL}v_n + \varepsilon_n^{RL}(t)$$
$$U_n^{LL}(t) = X_n^{LL}(t)\beta^{LL} + \alpha^{LL}v_n + \varepsilon_n^{LL}(t)$$

Short term plan and goal[7]



Conditional probability of acceptance of the lead gap[7]:

$$P_n\left(G_n^{lead\,TL}(t) > G_n^{lead\,TL,cr}(t) \middle| TL_t, v_n\right) = \phi\left[\frac{ln(G_n^{lead\,TL}(t)) - (X_n^{lead\,TL}(t)\beta^{lead} + \alpha^{lead} v_n)}{\sigma_{lead}}\right]$$

X.L. Wen et al, 2023[9]

- Identify the distinction between autonomous vehicles (AVs) discretionary lane-changing (DLC) and human-driven vehicles (HDVs) DLC maneuvers.
- DLC event definition:
  - move from current lane to TL
  - Longitudinal distance to following vehicle (FV) must be within 75 m
  - Speed must be bigger than 1 m/s
- Methology:
  - DLC duration
  - Effects of DLC on FV's behavior.
  - Extreme Value Theory (EVT) estimates Crash risks of DLCs
  - Non parametric ML Mars, models Gap acceptance
- More than 30% of crashes happen during lane change

Key contribution: crash analysis & crash evaluation

EVT issues:

- Not enough crash data between AVs and HDVs
- Crash data is a reactive approach
- The crash data unable to provide insights into particular driving behaviors

Block Maxima

Collects maxima in each interval.

randomly distributed Maximum Mn

Mn converges to a generalized extreme value (GEV) distribution as n -> $infinity$.

$$G(x) = \exp\{-\left[1 + \in \left(\frac{x-\mu}{\sigma}\right)\right]^{\frac{-1}{\in}}\}$$

Crash risk based on the GEV distribution

$$R = \Pr(Z \geq 0) = 1 - G(0)$$

Table of content

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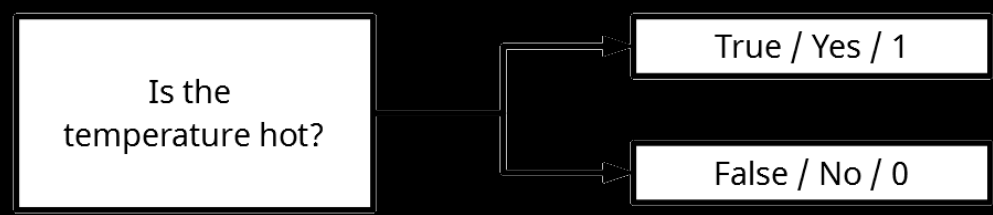
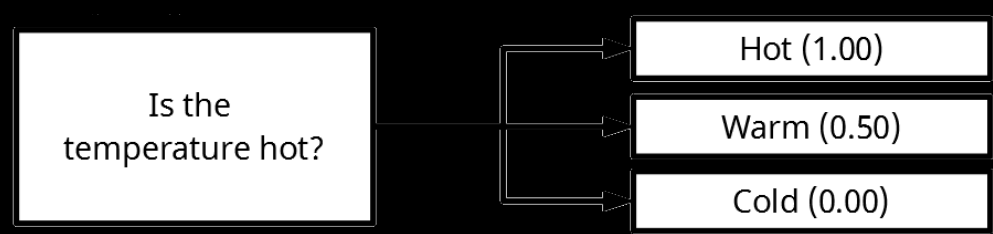
Fuzzy Logic

- Model logical reasoning, with vague or imprecise statements
- Model decides on:
  - The explanatory variables
    - The variables can be based on stages e.g. MLC or DLC, gap in target lane, and gap acceptance
  - Fuzzy sets and membership functions
    - The common membership functions are triangular, polynomial, trapezoidal, and gaussian
  - Rules sets
    - Lane changing to left or right
- The models are calibrated with an optimization algorithm
- Fuzzy rules are derived from the numerical data

Boolean Logic[11]



Fuzzy Logic[11]



McDonald, 1997[10]

- Mechanistic methods ignore perception & decision-making uncertainties in drivers behavior.
- Model:
  - Car Following (CF) model:
    - Fuzzy logic can predict drivers' behavior and psychological processes in interactions with preceding vehicles.
    - Premise variables: relative speed & distance divergence.
    - Each premise variable has a triangular membership function with a membership value.
    - Fuzzy system in CF Model parameters relative speed & headway changes based on it's speed & desired safe following distance.
  - Lane Changing model:
    - Driver Motivation on lane changing, varies with speed & safety benefits.
    - Two approaches:
      - Lane change to the offside (LCO)
        - Premise variables: opportunity & overtaking benefit
      - Lane change to the nearside (LCN)
        - Premise variables: pressure from the rear & gap satisfaction

Relative Speed (DV)		Distance Divergence (DSSD)		Driver Response (Acceleration Rate)
Opening Fast	(V1)	Much Too Far	(S1)	Strong Acceleration
Opening	(V2)	Too Far	(S2)	Light Acceleration
About Zero	(V3)	Satisfied	(S3)	No Action
Closing	(V4)	Too Close	(S4)	Light Deceleration
Closing Fast	(V5)	Much Too Close	(S5)	Strong Deceleration

Fuzzy Set for the Car Following model[10]

Overtaking Benefit		Opportunity		Intention of LCO
High	(OB1)	Good	(OP1)	High
Medium	(OB2)	Moderate	(OP2)	Medium
Low	(OB3)	Bad	(OP3)	Low

Fuzzy Set for the LCO[10]

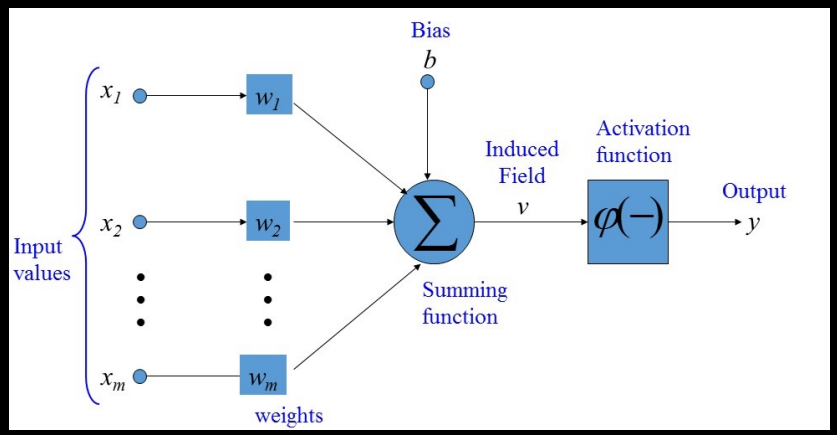
Pressure from Rear		Gap Satisfaction		Intention of LCN
High	(PR1)	High	(GS1)	High
Medium	(PR2)	Medium	(GS2)	Medium
Low	(PR3)	Low	(GS3)	Low

Fuzzy Set for the LCN[10]

Machine Learning prediction model for Lane Change

Artificial Neural Network (ANN)[2]

Structure of the ANN[12]:



- Input are environment variables (left and right lane density, exit distance, merge distance, driver satisfaction etc.)
- Input neuron → hidden layer →activation function
- Weights between different neurons → minimize error margin
- Bias value
- Output layer produces predicted probability of lane change to TL
- ANN is best in combination with ML

$$Y = f\left(\sum_{i=1}^{N} w_i x_i + b\right)$$

Y is the predicted probability, w_i weight, b bias value, x_i input value, and f is the activation[2]

Reason:

- Needs to train → being able mimic human driving to decrease amount of surprises on the road & behavior is consistent.
- Compared to mechanistic model the probabilistic return and does not give a predictive behavior.

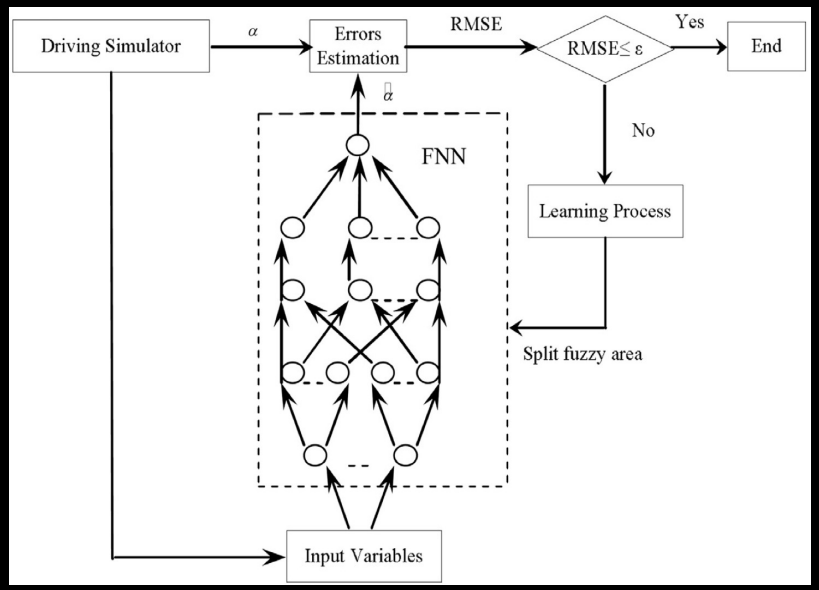
J. Tang, 2017[8]

- Predicts lane change through steering angle.
- Adaptive learning algorithm combined with the fuzzy neural network.
- The two key factors: the heading angles & vehicle acceleration.

Method[8]:
(1) Collects Data from the Driving Simulator
(2) The Fuzzy Neural Network (FNN) structure includes four sub-layers:

1. Input variables (distance between subject and lead vehicle, relative velocity, heading angle, acceleration),
2. Membership functions
3. Fuzzy reasoning mechanism
4. Output variable (steering wheel angles α)

(3) Evaluates Error estimation through the RMSE



- Tang approach includes new input variables (heading angle & vehicle acceleration)
- Still it is limited to predefined set of variables
- The solution is to find an adaptive mechanism to generate the list of the input and output variables, which could be rule based.

Table of content

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Conclusion

- We are in a transition time where AV's and HDV are operating on the roads, as well there are going to be vehicles that are connected to each other and vehicles that aren't.
- As evident from different approaches, there might not be a single approach that serves for all scenario.
  - A scenario based approach combing different parts from each model gives the best way forward. It may be required for Human drivers to take a step back and unlearn some decision approach to adapt to a AV integrated traffic flow.
- My take is that we are 4 to 5 years away till we see AVs and HDVs operating smoothly together.

- Eye opening experience to get to learn how to conduct research.
  - Having frustrating and rewarding moments.
  - It was definitely a worthwhile preparation before my Bachelor thesis.
  - I have more admiration to your work
    - Conducting research on your day to day basis
    - I want to especially thank Matteo supervisor on my project

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Current Example

Mercedes Benz Lane change assistance: https://youtu.be/wVOhUjrDuGQ?si=bglYpZUHJPmXgyhE
Tesla :https://www.youtube.com/watch?v=W5Eo5EEbt0s

AI & maximization approach → generalized in a non convex space, where it is a higher space dimension → use a continuous optimization function

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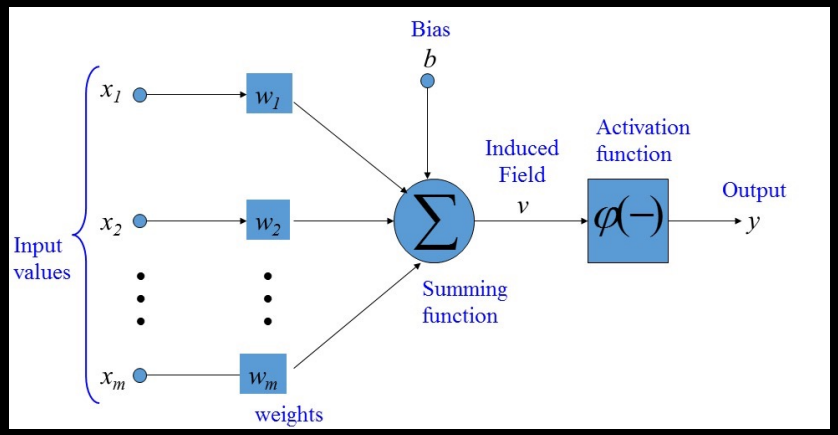
Machine Learning prediction model for Lane Change

Artificial Neural Network (ANN)



- Input are environment variables (left and right lane density, exit distance, merge distance, driver satisfaction etc.)
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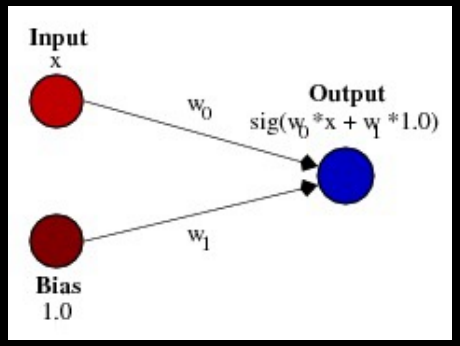
Y is the predicted probability, w_i weight, b bias value, x_i input value, and f is the activation



Reason:

- Needs to train → being able mimic human driving to decrease amount of surprises on the road & behavior is consistent.
- Compared to mechanistic model the probabilistic return and does not give a predictive behavior.

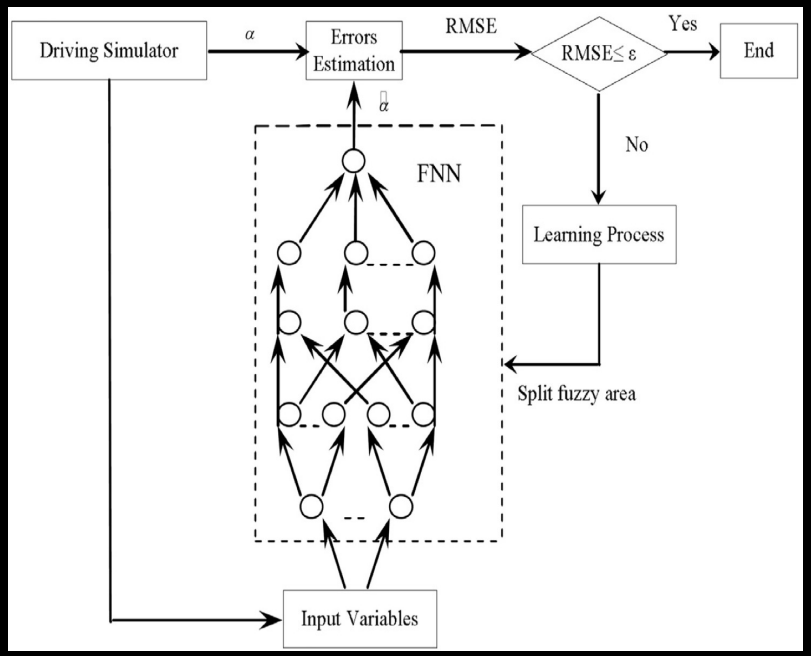
J. Tang et al, 2017

- Predicts lane change through steering angle.
- This approach makes use of the adaptive learning algorithm combined with the fuzzy neural network.
- The two key factors in this method are the heading angles and vehicle acceleration.
- Its weaknesses:
  - lack of comprehensiveness
  - incapability to incorporate adaptive learning mechanisms
  - limited use of indicators or factors in fuzzy input variables.
- Solution:(1) Create the rule base and inference mechanism, establish the input and output variables, and develop the FNN model. (2) With the use of prediction errors, the adaptive learning process can adjust the structure of fuzzy membership functions and enhance the fuzzy reasoning process. (3) The output variable the driving steering angle is determined through the input variables: vehicle speed, acceleration, heading angles, and distance from the front vehicle in the horizontal axis and vertical axis.

Lane changing prediction
System

(1) The data collected from the driving simulator included vehicle parameters, vehicle locations, instantaneous velocity, acceleration, heading, and steering wheel angles.

(2) The Fuzzy Neural Network (FNN) structure includes four sub-layers:

1. Input variables (distance between subject and lead vehicle, relative velocity, heading, acceleration),
2. Membership functions
3. Fuzzy reasoning mechanism
4. Output variable (steering wheel angles α)

(3) The adaptive learning algorithm splits fuzzy membership functions based on prediction errors, updates fuzzy rules, and calculates the new Root Mean Square Error (RMSE). If the RMSE is below a threshold, the prediction process stops, and the learning process continues until satisfactory results are achieved.